Protokoll des KI-Agenten (RE 10)

Gesichert aus dem Verlauf des Subagenten am 25.10.2026 (Zeiten UTC, Ortszeit Zürich = UTC + 1). Pfade gekürzt: `…` = Arbeitsordner des Versuchs (`ki-versuch/` im temporären Verzeichnis der Sitzung). Der Ordner `app/` dieses Repos ist eine unveränderte Kopie von `…/app/`.

Rückfragen

Der Agent konnte im Versuch keine Rückfragen stellen (einmaliger Auftrag, kein Dialog). Er hat keine Frage an uns gerichtet. Seine 15 Fragen an die Auftraggeberin (F-01 bis F-15) und die 32 Annahmen (A-01 bis A-32) stehen vollständig in `app/ANNAHMEN.md`.

Werkzeugaufrufe

35 Aufrufe in 28 Minuten. Die Spezifikation wurde einmal und vollständig gelesen (Aufruf 2, Option «ganze Datei»); andere Dateien hat der Agent nicht gelesen. Aufrufe 17–20 laden Prüfwerkzeuge aus dem Internet (playwright-core, Systembibliotheken für Chromium) in einen Hilfsordner `.tools/`, den der Agent am Ende wieder gelöscht hat. Aufrufe 22–30 sind Bildschirmfotos seines eigenen Testlaufs, die er angesehen hat.

#	Zeit	Werkzeug	Aufruf (erste Zeile)
1	14:52	Bash	`cd …/ && ls -la input app 2>&1; wc -c input/*`
2	14:52	Read	`…/input/anforderungsspezifikation-arenaone.md`
3	14:55	Bash	`mkdir -p …/app/{js,css}`
4	14:55	Write	`…/app/js/util.js`
5	14:56	Write	`…/app/js/store.js`
6	15:01	Write	`…/app/js/logic.js`
7	15:01	Bash	`cd …/app/js && python3 - <<'EOF'`
8	15:02	Write	`…/app/js/seed.js`
9	15:03	Bash	`cd …/app/js && python3 - <<'EOF'`
10	15:04	Write	`…/app/js/ui.js`
11	15:06	Write	`…/app/js/views-kunde.js`
12	15:10	Write	`…/app/js/views-betrieb.js`
13	15:11	Write	`…/app/index.html`
14	15:12	Write	`…/app/css/style.css`
15	15:13	Bash	`mkdir -p …/app/tests && cat > …/app/tests/logik-test.js <<'EOF'`
16	15:14	Bash	`cd …/app && sed -i "/const s = U.addMin(U.now().slice(0, 14)/d" tests/logik-test.js && node tests/lo`
17	15:14	Bash	`cd …/app && mkdir -p .tools && cd .tools && npm init -y >/dev/null && npm i playwright-core@1.56 --s`
18	15:15	Bash	`cd …/app/.tools && echo '{"name":"tools","version":"1.0.0","private":true}' > package.json && npm i`
19	15:15	Bash	`cd …/app/.tools && mkdir -p debs libs && cd debs && apt-get download libnspr4 libnss3 libasound2t64`
20	15:15	Bash	`cd …/app/.tools && export LD_LIBRARY_PATH=$PWD/libs/usr/lib/x86_64-linux-gnu && ldd ~/.cache/ms-play`
21	15:15	Bash	`cd …/app/.tools && mkdir -p shots && cat > smoke.js <<'EOF'`
22	15:16	Read	`…/app/.tools/shots/01-verfuegbarkeit-anonym.png`
23	15:16	Read	`…/app/.tools/shots/06-belegung-empfang.png`
24	15:16	Read	`…/app/.tools/shots/07-erfassen.png`
25	15:16	Read	`…/app/.tools/shots/10-serien.png`
26	15:16	Read	`…/app/.tools/shots/12-sperre-betroffen.png`
27	15:16	Read	`…/app/.tools/shots/16-auswertung.png`
28	15:16	Read	`…/app/.tools/shots/22-mobile-meine.png`
29	15:16	Bash	`cd …/app && python3 - <<'EOF'`

#	Zeit	Werkzeug	Aufruf (erste Zeile)
30	15:17	Read	`…/app/.tools/shots/16-auswertung.png`
31	15:17	Bash	`cd …/app && python3 - <<'EOF'`
32	15:18	Write	`…/app/ANNAHMEN.md`
33	15:19	Write	`…/app/README.md`
34	15:19	Bash	`cd …/app/.tools && LD_LIBRARY_PATH=$PWD/libs/usr/lib/x86_64-linux-gnu timeout 170 node smoke.js 2>&1`
35	15:20	SubagentHandback	`Abschlussbericht an den Auftraggeber`

Abschlussbericht des Agenten (wörtlich)

Der Agent hat seinen Bericht auf Englisch geschrieben, obwohl der Auftrag deutsch war; App, README und ANNAHMEN.md sind deutsch. Unverändert übernommen:

> I built a working ArenaOne app from the spec: a static web app (HTML, CSS, plain JavaScript) with no build step, no server and no external libraries. Data lives in localStorage and demo data is created on first start. It opens directly via `file://` and should run unchanged on GitHub Pages, but I haven't deployed it there. My 28 Node tests of the logic pass, and a run in headless Chromium went through every role's main flow with no JS errors. At 360 px width there is no horizontal scrolling.
>
> Folder: .../app/ (`index.html`, `css/`, `js/`, `tests/logik-test.js`, `README.md`, `ANNAHMEN.md`).
>
> **What was built.** Demo logins are listed on the login page. Customers use `demo1234`; staff use `arenaone-demo` plus a simulated second factor.
>
> - **Customers:** availability by date and sport, also without login (FA-01); online booking with rule checks and free alternatives on a clash (FA-04, FA-06); moving and cancelling, with the cancellation deadline and the "kurzfristig" flag (FA-09 to FA-12).
> - **Waitlist:** everyone on it is notified at once and whoever confirms first gets the slot (FA-13 to FA-15).
> - **Empfang:** day/week timeline of all facilities (FA-02, FA-20); booking on a customer's behalf with channel and quick customer search (FA-05, FA-11); customer accounts with several authorised people (FA-07, FA-08).
> - **Series:** request, conflict list, confirm in full or without the conflicting dates, reject with a reason, cancel or move single dates (FA-16 to FA-19).
> - **Closures:** a decision per affected booking is required before saving, and those affected are informed (FA-23, FA-27, FA-28).
> - **Notifications:** confirmations, reminders and the SMS option (FA-21, FA-22, FA-35). Nothing is really sent; messages go to a "Nachrichten" page.
> - **Geschäftsleitung:** customer groups, booking rules, quotas with the PA-10 limit, all parameters PA-01 to PA-10, and analyses by facility, sport, customer group, weekday and hour, plus cancellations, with CSV export (FA-24 to FA-26, FA-29 to FA-31).
> - **Administration:** facilities and opening hours, users and roles, import of old lists from CSV with a dry run and a conflict list, data access request, deletion of inactive accounts, and backup by download/upload (FA-32 to FA-34, QA-11, QA-12).

- **Cross-cutting:** the permission matrix enforced in the UI and in the logic (QA-13), an immutable change log (QA-15), data minimisation (QA-10), German throughout (QA-04), a 7-day printable list (QA-06), and an open second tab updates immediately (FA-03).

Not implemented, or only partly, and why:

- **QA-05, QA-07, QA-08, QA-09 (uptime, load, response times):** these need a shared server, which the brief ruled out. "Several sessions" only means several tabs in the same browser.
- **QA-16:** encryption in transit would come from GitHub Pages' HTTPS. Backup is manual JSON download/upload, with no daily automatic run.
- **QA-14:** login and second factor are simulated and passwords are stored in plain text in the browser. This is not real security.
- **FA-21 to FA-23, FA-35:** no real email or SMS, only the simulated outbox.
- **QA-01, QA-02:** the screens are built for fast use but not measured with test users.
- **WO-01 to WO-05:** left out on purpose, as the spec marks them "Won't" for v1.

Assumptions logged in `ANNAHMEN.md`: 32 (A-01 to A-32), plus 15 questions for the client (F-01 to F-15). The ones that most affect what the app does:

- **FA-26 example vs. PA-10:** the example quota "Mo–Fr 18–21 Uhr" takes 60 % of evening hours, which breaks the 50 % limit. I applied the limit strictly, so that example is rejected.
- **Hall parts:** a booked part blocks the whole hall and the other way round (OP-13 is open).
- **Booking lead time:** it applies to single bookings only, not to series.
- **Empfang:** may override customer-group rules, and every override is logged.
- **"Betrieb" role:** FA-27 mentions it but the QA-13 matrix doesn't, so I added a role that can only set closures.
- **Deletion vs. change log:** QA-11 (delete inactive accounts) conflicts with QA-15 (log entries can't be changed). On deletion, only the user name in the log is replaced with "gelöschtes Konto".
- **Admin and availability:** the matrix bars the admin from seeing availability even though it is public without login. I followed the matrix as written.

I used only the spec. To run the browser check I temporarily installed playwright-core and Chromium libraries into the work folder and removed them afterwards, so only the app files are left.